

Course syllabus for

Academic year 2022/2023 >

FUF040 - Quantum physics

Kvantfysik

Course syllabus adopted 2022-02-02 by Head of Programme (or corresponding)

Owner: [TKTFY](#)**6,0 Credits****Grading:** TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail**Education cycle:** First-cycle**Main field of study:** Chemical Engineering with Engineering Physics, Engineering Physics**Department:** 16 - PHYSICS**Teaching language:** Swedish**Application code:** 57114**Open for exchange students:** No**Block schedule:** [C](#)

Maximum participants: 170

Credit distribution

Module				Sp1	Sp2	Sp3	Sp4	Summer course 	No Sp	Examination dates 
0106	Written and oral assignments	6,0 c	Grading: TH	6,0 c						

In programs

[TKGBS GLOBAL SYSTEMS ENGINEERING, Year 3 \(elective\)](#)[TKKEF CHEMICAL ENGINEERING WITH ENGINEERING PHYSICS, Year 3 \(compulsory\)](#)[TKTFY ENGINEERING PHYSICS, Year 3 \(compulsory\)](#)

Examiner:

[Tom Blackburn](#)[Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

You should have studied the basic courses in mathematics and mechanics.

Aim

The aims of this first course are that participants should become familiar with the fundamental principles and experimental necessity of quantum mechanics, as well as develop the basic knowledge that will be built upon in subsequent courses. Participants will acquire the mathematical and conceptual tools needed to describe physical systems quantum-mechanically, and then apply these tools to generate predictions of practically relevant quantum-mechanical phenomena. It also aims to bring out the connections between quantum-mechanical concepts and classical physics, chemistry, and technological applications. The emphasis is on problem solving.

Learning outcomes (after completion of the course the student should be able to)

- Describe quantum-mechanical systems using quantum states, wavefunctions and probability amplitudes.
- Predict particle motion in one-dimensional and three-dimensional (spherically symmetric) potentials, including the "particle in a box", the harmonic oscillator, and the hydrogen atom.
- Explain quantum-mechanical phenomena in terms of wavefunction collapse, interference and the uncertainty principle.

Content

1. Experimental background: why do we need quantum mechanics?
2. Probability and probability amplitudes
3. States and wavefunctions
4. Position and momentum operators, the Hamiltonian
5. Measurement and expectation values
6. Time-independent and time-dependent Schrödinger equations
7. Free particles
8. One-dimensional potentials: bound states, scattering and tunnelling
9. The harmonic oscillator
10. Time-independent perturbation theory
11. Rotations, angular momentum and spin
12. The hydrogen atom

Organisation

Lectures and problem sessions.

Literature

"Introduction to Quantum Mechanics" by David J. Griffiths

Examination including compulsory elements

Hand-in problems (20% x 3), compulsory written test (40%), and optional oral exam. To pass you must achieve at least 40% of the total points available both in the hand-ins and test, separately.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.